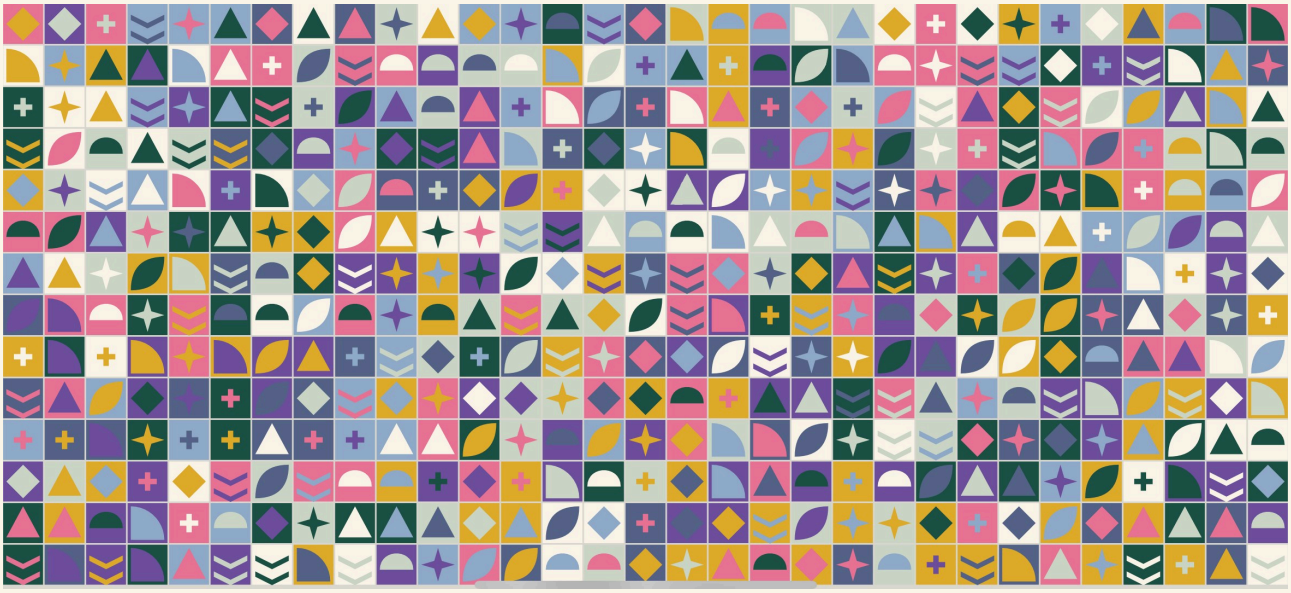


Speculae

PHASERS · DELAY · FOLD

created by k.odk

01 · THE IDEA

Two feedback oscillators

Speculae is an instrument made of two phasers, a bucket-brigade style delay and a fold of two sines, patched into each other so that each one feeds the next.

Nothing in it is a sample. Every sound comes from the loops themselves — a phaser fed back on itself rings at its own pitch, the delay turns it into a comb, the fold bends two sines into each other and pushes both phasers around. Whether they settle, beat or fall apart depends on how they are wired.

The screen is the instrument. It is one printed sheet of cells. Each block of cells is a part, and draws what that part is doing — the patterns swell, run and change colour with the sound. You play it by touching it: drag across for pitch or time, up and down for feedback, tap and hold to change how it is wired.

Underneath lies a mosaic of whole settings. Move across it and the sound mixes between them. Left alone, Speculae moves across the mosaic by itself; put a finger on the glass and it is played by hand, and nothing changes on its own.

Too much to learn? Skip it. Drag slowly across the mosaic in the middle until something catches your ear, then hold your finger still on that spot for a moment — the sound is kept there as a memory. If you get lost, the key at the very bottom left brings back the starting sound, the same every time.

Contents

02 Requirements

03 The screen at a glance

04 The top row — phasers, delay, fold

05 The mosaic — moving, circuits, fine control

06 Memory and presets

07 Thunor — a second instrument

08 Cafe — playing on Ciat-Lonbarde Cafes

09 Versions and support

What it needs

Speculae runs every loop at audio rate, sample by sample, while it draws the whole sheet thirty times a second.

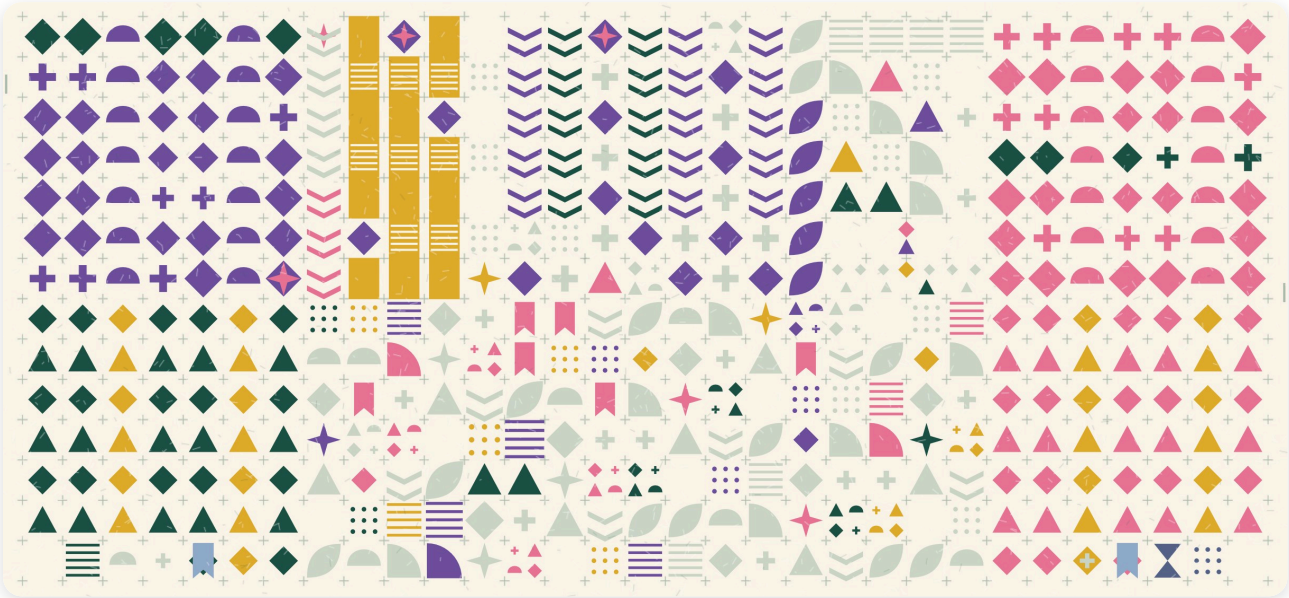
System

IOS	iOS 18.0 or later.
DEVICE	iPhone, played in landscape. The whole screen is the instrument; there are no panels to open except the manual.
RECOMMENDED	iPhone 13 or newer. Speculae is developed and tested on an iPhone 13.
BACKGROUND	It keeps playing while you switch apps or lock the screen, the way an instrument should. Stop it from the lock screen or Control Centre, or close the app.
BLUETOOTH	Only for Cafe (section 08), and only once you switch Cafe on in the manual. Until then the app never asks for it.
PRIVACY	Nothing is collected and nothing leaves the device, except the sound sent to your own Cafes over Bluetooth.

About AUv3

Speculae is a **standalone app**, not an AUv3 plug-in. The landscape, the mosaic and the memories are the instrument, and they do not fit inside a host's plug-in window.

One sheet, three bands FRONT



The top row

From left to right: **phaser a** (the large block of signs), the narrow **filter** column, the **delay**, the **fold**, the narrow **flange** column, the **patch** block (who goes into whom), and **phaser b**. Each draws what it is doing. Section 04.

The mosaic

The wide field below. Its points are whole settings; a small drop moves among them and mixes them. Section 05.

The edges

Slide along the very left or right edge of the screen: the pitch of phaser a or b.

The bottom row

clear	Far left (green bars). Every memory let go, back to the starting sound.
LFO	Beside it. The fold's two sines: both at pitch · the first an LFO · both LFOs.
presets	Two slots after it, two more at the right. Section 06.
Thunor	The hourglass. Section 07.
manual	Far right (nine dots). The guide inside the app, with Cafe at its top.

Phasers, delay, fold

Every block answers to the same three gestures: drag to move its two main values, tap for its next setting, hold for the other one.

phaser a · b	**Drag** across = pitch, up and down = feedback. **Tap**: the next stage — 2 · 4 · 6 · mix. **Hold**: notch ↔ peak (the inks change places). Its block swells with its sound; the lit row is its pitch, how many signs are whole is its feedback.
filter · flange	The narrow columns either side of the delay and the fold. **Drag** up and down.
delay	The bucket-brigade style delay. **Drag** across = delay, up and down = feedback — zero in the middle, below it the sound comes back inverted. **Tap**: where its feedback is taken from. **Hold**: which tap goes out.
fold	Two sines, the first folded over itself, shaped by the second, synced by its own output. **Drag** across = frequency, up and down = fold. **Tap**: the next ratio (1 · 1.5 · 2 · 3 · 4 · ½). **Hold**: its sync feedback. The LFO key at the bottom slows one or both sines down to LFOs.
patch	Who goes into whom: rows a · b · delay · fold, columns a · b · delay; a fuller sign carries more. Below it, the two ratios as whole numbers — a to b, and the fold's two sines — flash when they come round together. While Thunor sounds, this block is Thunor's (section 07).
beads	The small stars at the blocks' corners. **Pull** one onto another block to join them; do it again to part them.

Whole settings, mixed by place

The points on the mosaic are whole settings. The drop mixes them by where it is: near a point, the sound is that point.

drag	Move the drop.
tap	Glide there — or to the point or memory under the finger.
hold	Keep the sound as it is now, here — a memory (section 06).
playing	While a finger is on the glass it is played by hand: nothing changes by itself, and the pale cells blush.

Two fingers — fine control

Hold one finger on a part of the mosaic and move a second finger: that part's two values, finely — a fifth of the usual reach. The held part rings.

LEFT, ABOVE	Phaser a — pitch, feedback.
LEFT, BELOW	Phaser b — pitch, feedback.
MIDDLE	The fold — frequency, depth.
RIGHT, ABOVE	The delay — time, feedback.
RIGHT, BELOW	The delay — filter, flange.

Double tap — circuits and modulations

In the middle: the next circuit — as it starts · in a row · crossed · the delay first · loops. The lines between the phasers, the delay and the fold are laid anew.

On a side: that part's next modulation (none · one · the other), its colour changing with it — a's pitch by the delay / by b; b's by the delay / by a; the delay's time by a / by b; its filter by b / by a.

Keeping what you found

Two ways to keep a sound: a memory, kept at a place on the mosaic, and a preset, which keeps everything at once.

Memories

hold	On the mosaic, a moment: the sound as it is now is kept there — a rose bookmark. Hold on a bookmark to let it go.
tap	On a bookmark: the sound brought back exactly as it was kept — every value, key and line, and Thunor too — and left alone ten seconds before anything moves by itself again.
listening	Once memories are kept, the mix is not by distance alone. A small network learns the mosaic from the points and the memories, and listens: the mix leans toward the memories that sounded like now.
clear	The key at the bottom left lets every memory go, and resets to the starting sound.

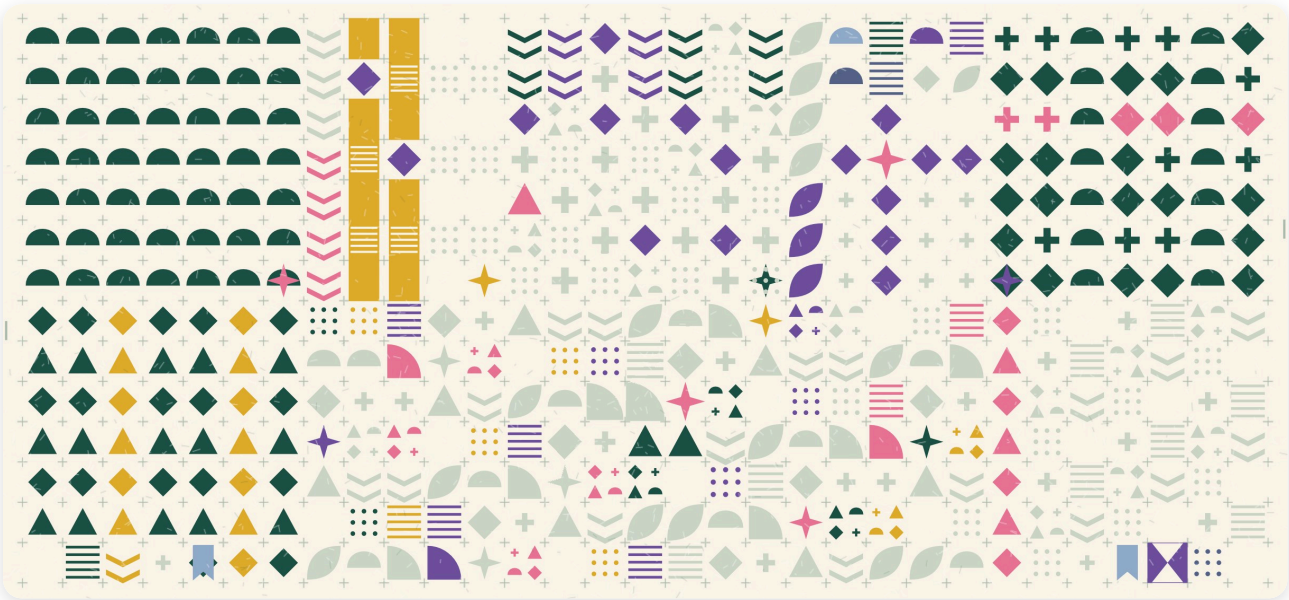
Presets

Four slots on the bottom row: two beside the LFO key at the left, two beside the hourglass at the right. A blue bookmark marks a slot that holds a preset; an empty slot is a faint cross.

hold	Keep everything as it is now in that slot: the memories, the points on the mosaic, the sound and its lines, the circuit and modulations, the fold's LFOs and Thunor.
tap	Bring a kept slot back whole. It is left alone ten seconds before anything moves by itself.
double tap	Clear that slot.

Memories and presets live in the app on your iPhone. Deleting the app deletes them.

A second instrument HOURGLASS



Thunor is the Thursday loop: two phasers and a delay in one ring, fed back on itself and swept by tent maps instead of random numbers — the same settings always take the same path. It lives in the hourglass key at the bottom right, beside the manual.

The hourglass — three states

off	A faint hourglass.
on its own	Violet. Thunor sounds alongside Speculae, straight to the output.
inside	Rose. Thunor is taken onto the ex / mic bus, so it goes wherever the lines send that — through the phasers or the delay.

Tap to step: off → on its own → inside → off. It switches the moment your finger lands.

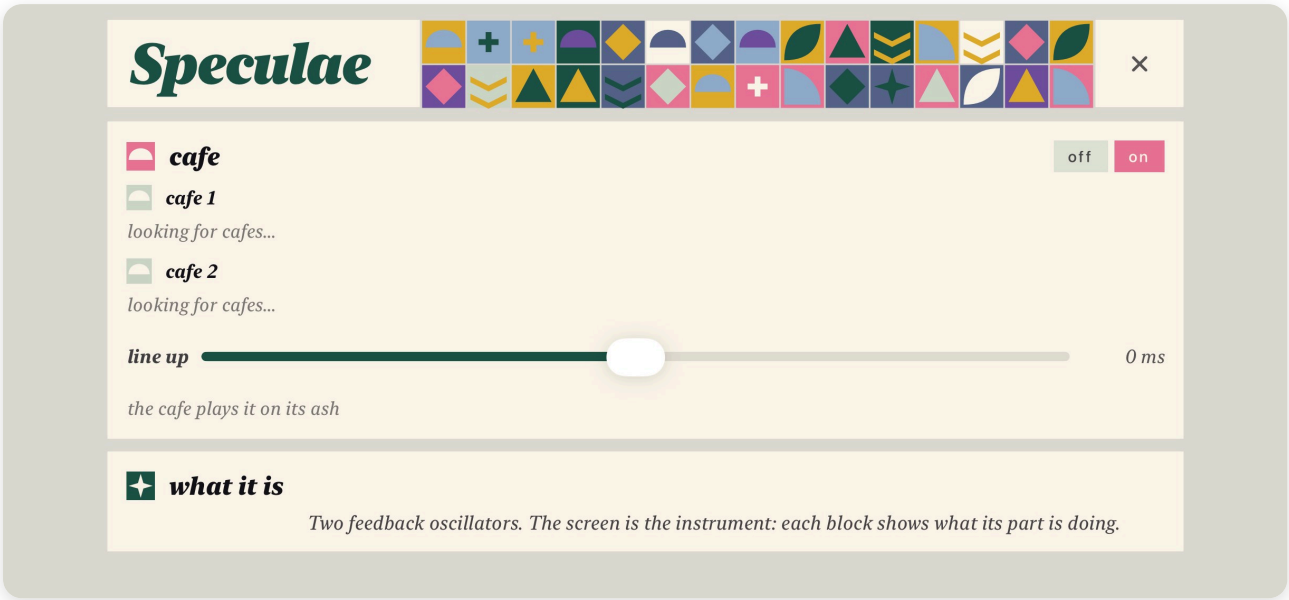
Its block

While Thunor sounds, the patch block upper right of the middle is its own. The two top rows are eight keys:

delay	single · dual (a second delay, four octaves up) · feed (the two in series, with a quiet sine fed in)
freq	the phasers' pitch, four steps
phase a · b	notch · peak · dual
shift a · b	360° · 720° · 1080°
pulse · drift	a grain on every high step of the map · the settings wander

The rows under the keys are its pad: across, the loop's time; up, its feedback. Past the hatched rows the loop sounds by itself. **Hold** the pad for a kick into the loop.

Playing on Ciat-Lonbarde Cafes



Up to two Ciat-Lonbarde **Cafes** can each play one of Speculae's outputs on their ASH. The Cafe needs the Apple Pi BLE firmware with the **speculae** preset. The sound goes over Bluetooth; the Cafe keeps a short buffer so a late packet does not break it, and while a Cafe sounds, the iPhone's own speakers play that much later too, so both line up.

Setting it up

on · off	At the top of the manual (the nine dots, bottom right). On, it looks for Cafes nearby.
cafe 1 · 2	Tap a Cafe's name to pair it to that slot. The × lets it go.
output	Choose what that Cafe plays.
line up	A fine trim (±150 ms) of how late the speakers play, until the two sound together.

The outputs

a · b	either phaser
delay l · delay r	either side of the delay
fold	the fold
out l · out r	either side of the output
thunor	the second instrument
a + b · delay	both phasers · both sides of the delay
fold + thunor	the two together
mix · ex / mic	the whole output · the ex / mic bus

Versions

1.0.0	First release. Two phasers, the delay and the fold on one sheet; the mosaic, memories and five circuits; the fold's LFO key; four presets; Thunor, the second instrument; Cafe.

Support

Something wrong, or something missing? Write to **kodk.dev@gmail.com**. Please include your device model, your iOS version, and what you were doing when it happened.

NOTHING PLAYS	Check the silent switch and the volume first. If it has wandered somewhere quiet, the key at the bottom left brings back the starting sound.
IT CHANGES BY ITSELF	That is deliberate: left alone, it moves across the mosaic on its own. A finger on the glass, a memory or a preset brought back keeps it still.
CAFE IS SILENT	Check the Cafe is on the speculae preset, that it shows ready in the manual, and that its output is not off.

Speculae is made by k.odk. Ciat-Lonbarde and Cafe belong to their makers; Speculae is not affiliated with them.